Analog Output MAPS Template (Instrumentation)

The Analog Output MAPS template is designed to control an analog that is connected to an analog control unit.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is available in three models – advanced, standard and basic, which provide respectively lower levels of control over the analog output.

This document describes the Advanced Analog Output MAPS Template and also defines the level of functionality provided by the standard and basic models. So that by reading this document you can decide which model best suits your needs. You should consider the following factors in your choice:

- the importance of the analog being controlled and
- the budget of the customer

Since the more complex the model, the greater the required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Note: The version 2 (v2) Analog Output MAPS template, communicates Floating point values instead of Integer values to the driver. The High-High and Low-Low Alarms does not latch as does not set the FAULT active. The Fault Active is only when the signal is out of range, meaning over or under range.

Advanced Analog Output Functional Philosophy

The analog output is scaled and alarmed according to the setpoints from the Setup faceplate.

Field Outputs

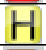
- Analog Output

PLC interlocks

- High-high Alarm
- High Alarm
- Low Alarm
- Low-low Alarm

Simulation mode - will disable the physical analog output and simulate an analog in the PLC program that is controlled from the faceplate. Simulation mode is used to test the process logic without receiving the physical output itself. It is very useful in testing, commissioning and maintenance.

Typical Faceplate Graphics for the Advanced Analog Output MAPS Template

Analog Output	
	
	Control/Tag name
	Analog Indication Value
	Fault active
	Manual mode
	High-High and Low-Low Alarm Active (Version 2 only)
	High and Low Alarm Active (Version 2 only)

Signal Description		**Agent Type**	**Advanced**	**Standard**	**Basic**
SCADA Control Words:					
SSCL 0	Control Word	**Marshal**	x	x	x
SSCL 1	High-high Alarm	**Analog**	x	x	
SSCL 2	High Alarm	**Analog**	x	x	
SSCL 3	Low Alarm	**Analog**	x	x	
SSCL 4	Low-low Alarm	**Analog**	x	x	
SSCL 5	Scale Min In Setpoint (Raw Min)	**Analog**	x		
SSCL 6	Scale Max In Setpoint (Raw Max)	**Analog**	x		
SSCL 7	Scale Min Out Setpoint (Eng. Min)	**Analog**	x		
SSCL 8	Scale Max Out Setpoint (Eng. Max)	**Analog**	x		
SSCL 9	Simulation Value	**Analog**	x	x	x
SSCL 10	Alarm Hysteresis	**Analog**	x	x	

Signal Description		**Agent Type**	**Advanced**	**Standard**	**Basic**
SCADA Status Words:					
SSSL 0	Status Word	**Marshal**	x	x	x
SSSL 1	Analog Output Process Value (Scaled)	**Analog**	x	x	x
SSSL 2	Analog Output RAW Value	**Analog**	x		

Analog Output:				
AO 0	Analog Output for Analog Output	x	x	x

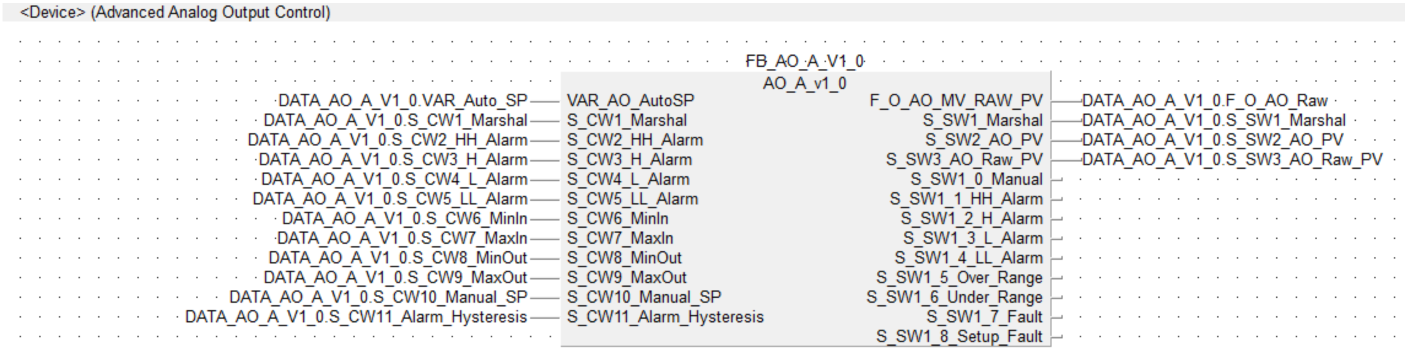
SCADA Control Word (Adroit Marshal Agent)		**Event Logged**	**Advanced**	**Standard**	**Basic**
Bit 0	Simulation	x	x	x	x
Bit 1	High-high AO Alarm Enabled	x	x	x	
Bit 2	High AO Alarm Enabled	x	x	x	
Bit 3	Low AO Alarm Enabled	x	x	x	
Bit 4	Low Low AO Alarm Enabled	x	x	x	
Bit 5	Reset Fault		x	x	x
Bit 6	Spare				
Bit 7	Spare				
Bit 8	Spare				
Bit 9	Spare				

Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

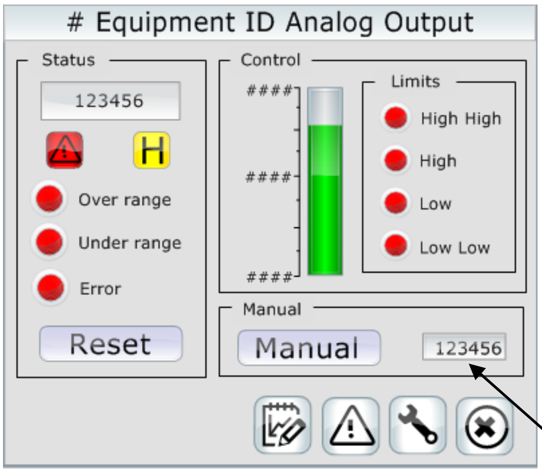
SCADA Status Word (Adroit Marshal Agent)		**Alarmed**	**Advanced**	**Standard**	**Basic**
Bit 0	Simulation On		**x**	**x**	**x**
Bit 1	High-high Alarm Active	**x**	**x**	**x**	
Bit 2	High Alarm Active	**x**	**x**	**x**	
Bit 3	Low Alarm Active	**x**	**x**	**x**	
Bit 4	Low-low Alarm Active	**x**	**x**	**x**	
Bit 5	Warning Active	**x**	**x**	**x**	
Bit 6	Alarm Active	**x**	**x**	**x**	
Bit 7	Alarm Fault Latch		**x**	**x**	**x**
Bit 8	Alarm Setup Fault		**x**	**x**	
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

Resources	**Advanced**	**Standard**	**Basic**
Digital Inputs	0	0	0
Digital Outputs	0	0	0
Analog Inputs	0	0	0
Analog Outputs	1	1	1
PLC Memory Steps	355	271	82
System Words Used	40	40	2
System Bits Used	71	63	51
SCADA SCAN Points	14	9	4

Template: Advanced Analog Output PLC Function Block

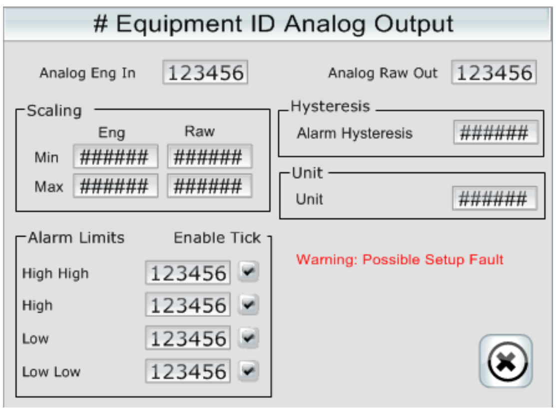


Template Graphic: Advanced Home Screen

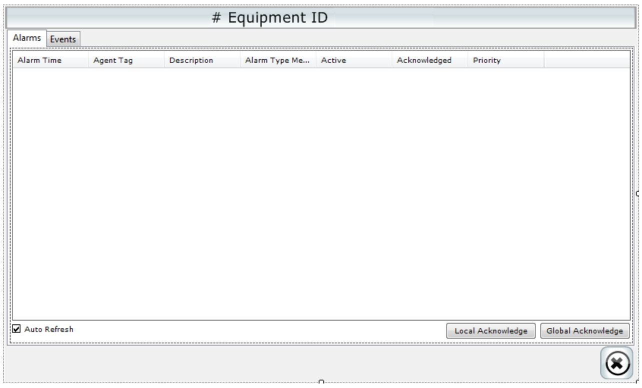


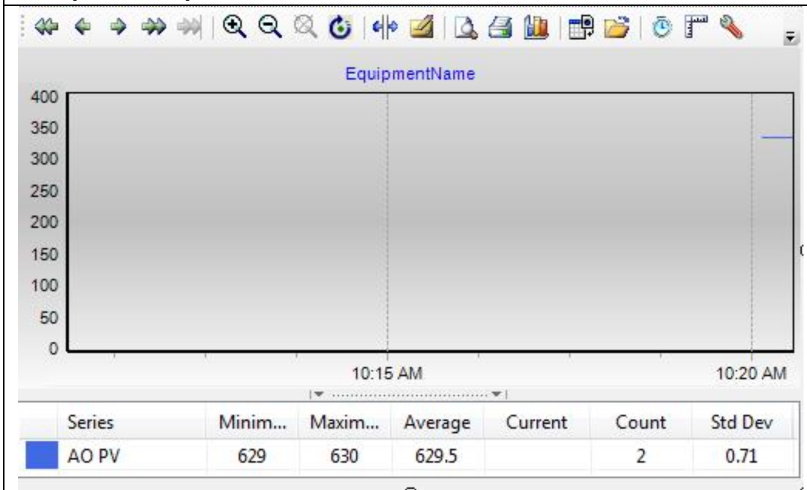
Control	Descriptions
# Equipment ID	Analog Output controlled name
High High	High-high alarm indication
High	High alarm indication
Low	Low alarm indication
Low Low	Low-low alarm indication
Over range	Analog over range alarm
Under range	Analog under range alarm
Error	Alarm fault latch
Reset	Reset the analog Output after a fault has been repaired
123456	Analog scaled value indication
Manual	Manual mode enable button
123456	Manual mode analog output default value
H	Manual indication
⚠	Alarm fault latch

Template Graphic: Advanced Setup Screen



Control	Descriptions
Analog Raw Out	Raw value scaled from input value for analog output
Analog Eng In	Value from PLC for analog
Min, Max Raw	Raw output values for scaling
Min, Max Scaled	Engineering input values for scaling
High High	High-high alarm setup value and enabling
High	High warning setup value and enabling
Low	Low warning setup value and enabling
Low Low	Low-low alarm setup value and enabling
Alarm Hysteresis	High and low warning auto reset hysteresis value. This is a value at which the warnings will reset should the process value decrease – in the case of the high level or increase - in the case of a low level
Unit	Unit displayed for the scaled analog

<table>
<tr><th>Template Graphic: Advanced Alarms & Events Screen</th><th>Control</th><th>Descriptions</th></tr>
<tr><td rowspan="4">
</td><td>Alarms</td><td>Display all the alarms filtered for the current Digital Output</td></tr>
<tr><td>Events</td><td>Display all the events filtered for the current Digital Output.</td></tr>
<tr><td>Local Acknowledge</td><td>Acknowledge all the alarms on locally on the current machine</td></tr>
<tr><td>Global Acknowledge</td><td>Acknowledge all the alarms on globally on all the machines.</td></tr>
</table>

<table>
<tr><th>Template Graphic: Advanced Trend Screen</th><th>Control</th><th>Descriptions</th></tr>
<tr><td rowspan="2">
</td><td>Equipment Name</td><td>Display the name of the current analog Output of the trend</td></tr>
<tr><td>AO PV</td><td>Display the level of the current level</td></tr>
</table>